

Integer Polynomials

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The guiding principle of today is the following:

Integer polynomials are *hard* to deal with.

There is plenty of theory regarding polynomials (even small-degree 2-variable polynomial equations) that already imply that there is much to be studied. This means that to polynomial problems on *olympiads* are either:

1. tractable using simple tools, or
2. extremely difficult.

Usually you'll only see the first kind appear. The second may also pop up if you venture into forbidden China TST territory.

WARNING: The remarks in this handout are rather difficult. If you don't understand them just skip them.

Acknowledgement: I've taken many problems, and a lot of inspiration, from the handouts on real / integer polynomials mentioned in the reading recommendations. Thank you to them!

1 Case Study: Elliptic Curves

This part can be skipped if you find it rather confusing.

For our purposes an elliptic curve will be an equation of the form

$$y^2 = x^3 + ax + b.$$

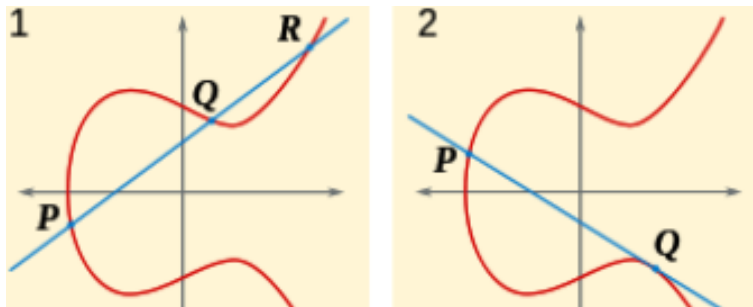
We may consider this equation over $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$, or modulo a prime p . Over $\mathbb{Q}, \mathbb{R}$, or $\mathbb{C}$, (or in $\mathbb{F}_q$ for $2, 3 \nmid q$) we may take *any irreducible* curve, and transform it into an elliptic curve of the above form using an affine changes of variables (i.e. $x \mapsto ax + b, y \mapsto cx + d$). So really the study of elliptic curves is the study of “most” cubic equations.

Given an elliptic curve E , what do we know about its rational points (i.e. rational solutions)? Here are two properties we do know:

1. if (x, y) is a solution, then so is $(x, -y)$;
2. given two solutions (x_1, y_1) and (x_2, y_2) , we may consider the line l passing through them. Then l intersects E at a third point, and this point has rational coordinates (by Vieta's). Here we count points with multiplicity.

These two properties define the *group law* of Elliptic curves:

Definition 1.1. Given an elliptic curve E over the rationals (i.e. we consider only its rational solutions), we may define an operation $+$: $E \times E \rightarrow E$ as follows. Given any two rational points $A = (a_1, a_2), B = (b_1, b_2) \in E$, define $C = (c_1, c_2) = A \oplus B$ to be the unique rational point on E satisfying that $(a_1, a_2), (b_1, b_2), (c_1, -c_2)$ are collinear. If $A = B$, instead require $(c_1, -c_2)$ to lie on the tangent to E at A , and require $(c_1, -c_2) \neq A$ if E does not have an inflection point at A .



In considering the group law, we consider the elliptic curve to also have a point at infinity. Formally, homogenize the elliptic curve as $y^2z = x^3 + axz^2 + z^3$; then the point at infinity is $[0 : 1 : 0]$.

Theorem 1.2. *The group law is well-defined, and makes E into a (commutative, i.e. abelian) group. That is, it has an identity element O , and satisfies that for any $A, B, C \in E$,*

1. $A + O = O + A = A$,
2. *there exists a D for any A such that $A + D = D + A = O$,*
3. $(A + B) + C = A + (B + C)$,
4. $A + B = B + A$.

Remark 1.3. Here is some high-level motivation for where this group law comes from. By algebraic geometry, an irreducible curve of degree d has genus $\frac{(d-1)(d-2)}{2}$. So an irreducible smooth cubic curve is a Riemann surface (i.e. orientable 2D shape) of genus 1, i.e. a torus. Such tori are (by Uniformization) all quotients of $\mathbb{C}$ by some lattice Λ . In fact, we may identify an elliptic curve E with such a torus $\mathbb{C}/\Lambda$ using an explicit parameterization (that is, using the Weierstrass phi function $\wp$ and its derivative). Under this identification the group law just becomes addition in $\mathbb{C}/\Lambda$.

Elliptic curves, over a variety of fields, are still the focus of intense study today! Hence the study of elliptic curves is both very rich, and very difficult. The upshot for you is the following heuristic:

Cubic 2-variable diophantine equations on olympiads are usually reducible, because otherwise they would be intractable!

Here's an easy example.

Example 1.4. (Putnam 2006 B1) Show that the curve $x^3 + 3xy + y^3 = 1$ contains only one set of three distinct points, A, B , and C , which are the vertices of an equilateral triangle.

1. Can you factor the cubic?
2. Solve the problem.

1.1 Practice

Here are some more examples of cubic curves on contests:

1. Find all triples of positive integers (x, y, z) that satisfy the equation

$$2(x + y + z + 2xyz)^2 = (2xy + 2yz + 2zx + 1)^2 + 2023.$$

2. Solve in integers the equation

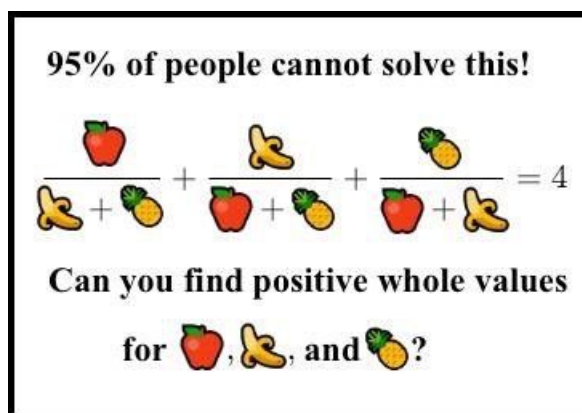
$$x^2 + xy + y^2 = \left(\frac{x+y}{3} + 1\right)^3.$$

3. Determine all pairs (x, y) of positive integers such that

$$\sqrt[3]{7x^2 - 13xy + 7y^2} = |x - y| + 1.$$

And here are some examples of actual elliptic curve problems:

1. Give an example of a pair of rational non-integer numbers a, b satisfying $a^3 + b^3 = 9$.
2. Solve the following puzzle.



2 Polynomial Basics

In terms of what you can do with integer polynomial equations, there are two main threads of ideas you can pursue:

1. you can use ideas that also apply to generic polynomials, or
2. you can use ideas specific to integer polynomials.

2.1 General Polynomial Practice

There are a lot of useful ideas for general polynomials, but the most useful one for you will be the *Factor Theorem*.

Theorem 2.1. Suppose a polynomial $p(x)$ has a root at c . Then $x - c$ is a factor of $p(x)$.

Proof. Note that $x - c$ divides $p(x) - p(c) = p(x)$ as a polynomial. □

Here's an example.

Example 2.2. (CMO 2013 1) Determine all polynomials $P(x)$ with real coefficients such that

$$(x+1)P(x-1) - (x-1)P(x)$$

is a constant polynomial.

1. Plug in $x = 0$ and $x = -1$.
2. Ugh, we have an annoying constant term. Shift P by a constant to get rid of it.
3. Factor terms and finish.

Here's a manipulation-based example.

Example 2.3. (HMMT 2017 A6) A polynomial P of degree 2015 satisfies the equation $P(n) = \frac{1}{n^2}$ for $n = 1, 2, \dots, 2016$. Find $2017P(2017)$.

1. To interpret the condition, multiply out n^2 . Factorize $n^2P(n) - 1$.
2. We have two unknowns remaining, so we need two equations. Find one by plugging in a certain n .
3. Now one of the coefficients of $n^2P(n) - 1$ is missing. Using this, get another equation.
4. Solve the problem.

Finally, here's a more olympiad-styled problem that shows the power of factorization.

Example 2.4. (Putnam 2007 B4) Let n be a positive integer. Find the number of pairs P, Q of polynomials with real coefficients such that

$$(P(X))^2 + (Q(X))^2 = X^{2n} + 1$$

and $\deg P < \deg Q$.

1. Factorize *both* sides.
2. Write $R(x) = Q(x) + iP(x)$. What are the possibilities for $R(x)$ and its complex conjugate?
3. Finish from here. As a detail, what are the possible leading terms of R ?

There are plenty more real polynomial ideas that I won't delve into today. Some other important ones:

1. looking at roots / estimating root sizes;
2. looking at symmetric sums;
3. applying Lagrange Interpolation;
4. taking derivatives.

Of course, polynomials are also continuous functions, so you can apply e.g. the intermediate value theorem. Also, the size of polynomials is very well-controlled.

For more information, see the following references:

1. Victor wrote a very expansive handout on real polynomials, which you can find on [his webpage](#).
2. OTIS has a D RealPoly unit that I think is quite helpful.
3. US college contests (especially [the Harvard-MIT Math Tournament \(HMMT\)](#) and [the Stanford Math Tournament](#)) have a lot of polynomial problems for some reason. They're all quite good.

2.2 An Important Divisibility

The most important idea for solving integer polynomial equations is the following generalization of the Factor theorem.

Theorem 2.5. *Consider a polynomial p with integer coefficients. Then for any $a, b \in \mathbb{Z}$, we have that*

$$a - b \mid p(a) - p(b).$$

Proof. For a, b integers, $\frac{a^n - b^n}{a - b}$ is an integer. □

This idea (combined with some usual real polynomial techniques) is enough for most integer polynomial problems!

Example 2.6. (CMO 2016 3) Find all polynomials $P(x)$ with integer coefficients such that $P(P(n) + n)$ is a prime number for infinitely many integers n .

1. Prove that $P(n) \mid P(P(n) + n)$.
2. Prove $P(n) \in \{\pm 1\}$ for infinitely many n .
3. Finish.

Example 2.7. (ELMO 2023 1) Let m be a positive integer. Find, in terms of m , all polynomials $P(x)$ with integer coefficients such that for every integer n , there exists an integer k such that $P(k) = n^m$.

1. Note that the result is true for $n = 0$. What does that tell you about $P(x)$?
2. Adjust $P(x)$ such that $P(0) = 0$. Now what happens when you plug in n a prime?
3. Apply pigeonhole and finish.

2.3 Further Properties of Integer Polynomials

Here are some other theorems that help structurally characterize polynomials.

Theorem 2.8. (Bézout) *Consider any two univariate polynomials $p(x), q(x)$ with rational (or real, or complex) coefficients. If $\gcd(p(x), q(x)) = r(x)$, then there exist polynomials $a(x), b(x)$ such that*

$$a(x)p(x) + b(x)q(x) = r(x).$$

As an exercise: why does Bézout *not* work for multivariate polynomials?

Theorem 2.9. (Schur) *Let P be a nonconstant polynomial with integer coefficients. Then there exist infinitely many primes p such that $p \mid P(n)$ for some integer n .*

1. Suppose $P(0) = 0$. Why are you done?
2. Now suppose $P(0) \neq 0$. Emulate a familiar argument by Euclid.

Here's an example.

Example 2.10. (Taiwan TST 2014 Round 1) For a fixed integer k , determine all polynomials $f(x)$ with integer coefficients such that $f(n)$ divides $(n!)^k$ for every positive integer n .

1. Suppose $p \mid f(n)$. Prove $p \leq n$.
2. Show that $x \mid f(x)$.
3. Let's now just say $f(x) = x^l g(x)$, where $x \nmid g(x)$. What can we say about l and g ?
4. Solve the problem.

Here's one more theorem that occasionally pops up in olympiads, but is generally very useful. It's a structural classification of how the v_p of polynomials behave.

Theorem 2.11. (*Hensel's Lemma*) Consider a polynomial f with integer coefficients, and a prime p . Consider an integer r satisfying $v_p(f(r)) = n > 0$ and $p \nmid f'(r)$. Then for every $m > n$, there exists a unique $s \bmod p^m$ such that

$$v_p(f(s)) = m \quad \text{and} \quad s \equiv r \bmod p^n.$$

2.4 A Warning

As you may have already noticed on this year's COMC, a polynomial which *takes integers to integers* does **NOT** have to have integer coefficients! For example, $\binom{x}{2}$ works just fine, and so does $\binom{x}{k}$ for any positive integer k . So how can we characterize the polynomials which send integers to integers?

Definition 2.12. For a polynomial $p(x)$, define Δp to be the polynomial $\Delta p(x) = p(x+1) - p(x)$.

Theorem 2.13. (*Newton's Finite Difference Formula*) Given any polynomial $P(x)$, define

$$c_k = \sum_{i=0}^k (-1)^i P(k-i) \binom{k}{i}.$$

Then

$$P(x) = \sum_{k \geq 0} c_k \binom{x}{k}. \tag{1}$$

1. Prove that $c_k = (\Delta^k P)(0)$.
2. We now induct on the degree of P . Prove the base case.
3. Prove that Equation (1) holds upon applying Δ .
4. Prove that $\Delta Q = 0$ iff Q is a constant.
5. Prove that the constants on both sides of Equation (1) match, and finish the proof.

2.5 Hard Example

Here's a difficult but very instructive example.

Example 2.14. (2011 N6) Let $P(x)$ and $Q(x)$ be two polynomials with integer coefficients, such that no nonconstant polynomial with rational coefficients divides both $P(x)$ and $Q(x)$. Suppose that for every positive integer n the integers $P(n)$ and $Q(n)$ are positive, and $2^{Q(n)} - 1$ divides $3^{P(n)} - 1$. Prove that $Q(x)$ is a constant polynomial.

1. First let's decode the coprimality condition. Prove that there exist integer polynomials $A(x), B(x)$, and an integer constant c , such that $A(x)P(x) + B(x)Q(x) = c$ always. This will be useful later.
2. Let's now work with the problem. Prove that for any prime p ,

$$\text{ord}_p(2) \mid Q(n) \implies \text{ord}_p(3) \mid P(n).$$

3. Show that for any sufficiently large integer n , we can take $n = \text{ord}_p(2)$. We'll soon want to take n a prime.
4. Consider any large prime q dividing some $Q(n)$; by Schur, infinitely many such primes exist. Prove that for some integer x_q , $q \mid Q(n)$ implies $x_q \mid P(n)$. Prove also that as q increases, we can choose the corresponding x_q to also become unbounded (i.e. for any constant C , there exist infinitely many large primes q dividing some $Q(n)$, such that the corresponding x_q are larger than C).

5. We'll now do two cases. Suppose infinitely many of the x_q are divisible by q . Show that then $q|c$ and finish.
6. Suppose infinitely many of the x_q are not divisible by q . Show that then q divides all residues of P , and finish.

3 Algebraic Numbers and Minimal Polynomials

When working with real polynomials, one useful idea is to consider its *roots*. In the real polynomial context it's mostly useful to consider how large its roots can be, and how this data interacts with the sizes of the coefficients of the polynomial.

We'll now find that we can develop a very precise theory for the roots of *integer* polynomials. We'll begin with some definitions.

Definition 3.1. An *algebraic number* is a complex number which is the root of some integer polynomial. We denote the set of algebraic numbers as $\overline{\mathbb{Q}}$.

Observe that all roots of rational-coefficient polynomials are also algebraic numbers, since we can just multiply out the denominators.

Definition 3.2. An *algebraic integer* is a complex number which is the root of some *monic* integer polynomial. We denote the set of algebraic integers as $\overline{\mathbb{Z}}$.

So all integers are algebraic integers, and so are all radicals $\sqrt[n]{n}$.

Example 3.3. Given any two distinct positive integers a, b , prove that $\sqrt{a} + \sqrt{b}$ is an algebraic integer.

Using a similar theme we may show that the sum (or product) of any two algebraic integers (or numbers) is also an algebraic integer (or number).

Theorem 3.4. *The algebraic integers are closed under addition, subtraction, and multiplication. (In higher math terms, algebraic integers form a ring).*

Proof. Suppose α is the root of a monic integer polynomial $P(x)$, whose set of roots is $\alpha = \alpha_1, \alpha_2, \dots, \alpha_n$. Similarly say β is the root of a monic integer polynomial $Q(x)$, whose roots are $\beta = \beta_1, \beta_2, \dots, \beta_m$. Consider the polynomial

$$R(x) = \prod_{i,j} (x - \alpha_i - \beta_j)$$

By elementary symmetric polynomial theory, given any monomial $\beta_1^{e_1} \dots \beta_m^{e_m}$, its coefficient in the polynomial above is symmetric in the α_i , and thus is in fact an integer. Applying symmetric polynomial theory to the β_i s now implies that the coefficients of $R(x)$ are integers. A similar proof implies that algebraic integers are closed under multiplication, and the algebraic integers are closed under additive inverses by taking $p(x) \mapsto p(-x)$. $\square$

Similarly we may prove the following:

Theorem 3.5. *The algebraic numbers are closed under addition, subtraction, multiplication, and division. (In higher math terms, algebraic numbers form a field).*

Proof. This time, we just need to show that the reciprocal of any nonzero algebraic number is an algebraic number. First observe that by replacing $p(x) \mapsto p(ax/b)$, we have that $\overline{\mathbb{Q}}$ is invariant under scaling. Now given any nonzero $\alpha \in \overline{\mathbb{Q}}$, we know α is the root of some polynomial $P(x)$, whose other roots are $\alpha_2, \dots, \alpha_n$. Then Vieta's tells us that $\alpha_2 \dots \alpha_n = (-1)^n P(0)/\alpha$, so $(-1)^n P(0)/\alpha$ and hence $1/\alpha$ are both algebraic numbers. $\square$

Now, a very useful property of algebraic numbers is that they each are associated with a unique polynomial.

Definition 3.6. An integer polynomial is *primitive* if the gcd of all its coefficients is 1.

Definition 3.7. The *minimal polynomial* of an algebraic number α is the unique primitive integer polynomial $P(x)$ of minimal degree such that $P(\alpha) = 0$ and the leading coefficient of $P(x)$ is positive.

Evidently the minimal polynomial is unique, as if two minimal polynomials P, Q existed for a given α , then some linear combination of P, Q would be nonzero but have α as a root, contradicting minimality. Observe also that the minimal polynomial must be irreducible.

Example 3.8. Compute the minimal polynomial of:

1. all integers;
2. all rationals;
3. all radicals $\sqrt{n}$;
4. the number $\sqrt{2} + \sqrt{3}$.

Theorem 3.9. Given an algebraic number α with minimal polynomial $P(x)$, all integer polynomials $Q(x)$ which have α as a root are divisible (as integer polynomials) by $P(x)$.

For this we'll need a lemma.

Lemma 3.10. (Gauss) A primitive integer polynomial is irreducible in $\mathbb{Z}[x]$ iff it is irreducible in $\mathbb{Q}[x]$.

Note that being reducible in $R[x]$ for R a ring (in this context, just $\mathbb{Z}$ or $\mathbb{Q}$) just means that you can be factored into two polynomials, both of which are also in $R[x]$ (i.e. have coefficients in R).

Proof. Take out constant integer factors of Q until it is primitive. First, by the division algorithm in $\mathbb{Q}[x]$ we may express $Q(x) = P(x)A(x) + B(x)$ for $A(x), B(x) \in \mathbb{Q}[x]$. Now plugging in α gives $B(\alpha) = 0$; by minimality $B(x) \equiv 0$. So Q is reducible in $\mathbb{Q}[x]$, hence by Gauss' lemma Q is reducible in $\mathbb{Z}[x]$, and in fact must have $P(x)$ as a factor, as desired. $\square$

So the minimal polynomial of α precisely determines the polynomials which have α as a root! Notably, the following lemma follows immediately from the above theory.

Lemma 3.11. An algebraic number is not an algebraic integer iff its minimal polynomial has leading coefficient of absolute value greater than 1.

The following follows immediately from this lemma:

Lemma 3.12. The only rationals which are algebraic integers are the integers. In other words, $\mathbb{Q} \cap \overline{\mathbb{Z}} = \mathbb{Z}$.

Proof. The minimal polynomial of p/q is $qx - p$. $\square$

Here are some problems which utilize the basic elements of this theory.

Example 3.13. (Niven) Consider a rational number q . Suppose $\cos(q\pi)$ is rational; then $\cos(q\pi) \in \{\pm 1, \pm 1/2, 0\}$.

1. Let $z = e^{iq\pi}$. Prove that z is an algebraic integer.
2. Show that $2\cos(q\pi)$ is also an algebraic integer, and finish.

Example 3.14. (Black Mop 2010) Consider $n > 2$ lamps placed at the vertices of a regular n -gon. Each move we may choose, for some divisor $d > 1$ of n , d lamps forming a regular d -gon, and toggle all of these lamps once. Suppose the lamps begin with one on and all others off. Is it possible to turn all lamps off after a finite number of moves?

1. Find an invariant in terms of the number of toggles of each lamp.
2. Reduce the problem to showing we cannot express $1/2$ as a sum of roots of unity.
3. Prove this statement using algebraic integer theory.

3.1 Exercises

Here are some exercises related to the theory of this section. (Some of these exercises may need a little more theory though, so be careful!) For more exercises and a lengthier discussion, see OTIS' D Intpoly unit.

1. Infinitely many equidistant parallel lines are drawn in the plane. A positive integer $n \geq 3$ is called *frameable* if it is possible to draw a regular polygon with n sides all whose vertices lie on these lines, and no line contains more than one vertex of the polygon. Find all frameable positive integers.
2. Let θ be an angle in the interval $(0, \pi/2)$. Given that $\cos \theta$ is irrational, and that $\cos k\theta$ and $\cos[(k+1)\theta]$ are both rational for some positive integer k , show that $\theta = \pi/6$.
3. A needle lies on the plane. One can rotate it 45° about either endpoint. Is it possible that after several rotations the needle returns to initial position with the endpoints interchanged?
4. Consider an algebraic integer α with minimal polynomial p . Show that if all the roots of p have absolute value 1, then α is a root of unity.
5. Let ω be a root of unity and f be a polynomial with integer coefficients. Show that if $|f(\omega)| = 1$, then $f(\omega)$ is also a root of unity.
6. Let $\omega = e^{2\pi i/5}$ be a primitive fifth root of unity. Prove that there do not exist integers a, b, c, d, k with $k > 1$ such that

$$(a + b\omega + c\omega^2 + d\omega^3)^k = 1 + \omega.$$

4 Final Thoughts

A recap of today on how to approach integer polynomial problems:

1. Try applying generic polynomial ideas.
2. Consider using $a - b \mid p(a) - p(b)$ a *lot*.
3. Polynomials have some structure you can exploit, but you shouldn't need anything crazy to solve olympiad problems. The theory gets really deep, so if you think you need anything crazy, that probably just means you may be overcomplicating the problem (or the problem is unreasonably hard)

Some reading recommendations:

1. For general polynomials, look at [Victor's handout last year](#), or OTIS' D RealPoly unit.
2. For integer polynomials, look at [Victor's handout last year](#), look at OTIS' D IntPoly unit.
 - (a) This latter unit is really hard to get a grasp on though if you don't know actual abstract algebra. For that I would recommend taking the geodesic in Artin's *Algebra* to the chapters about Finite Fields and Galois Theory. (I would not really recommend Napkin; this material is very structural and should be studied systematically, so Napkin is not really comprehensive enough).

5 Problems

5.1 Core Problems

Here are some polynomials that chiefly use the ideas in today's handout.

5.1.1 Easier

Here are some IMO 1-2 level problems.

1. Let f be a polynomial with positive integer coefficients. Prove that if n is a positive integer, then $f(n)$ divides $f(f(n) + 1)$ if and only if $n = 1$.
2. Given two polynomials P, Q with integer coefficients, suppose for infinitely many integers n we have that $Q(n) \mid P(n)$. Prove that $P(x)/Q(x)$ is a polynomial with rational coefficients. Does it have to have integer coefficients?
3. Let a, b , and c denote three distinct integers, and let P denote a polynomial having integer coefficients. Show that it is impossible that $P(a) = b$, $P(b) = c$, and $P(c) = a$.
4. Big Bird has a polynomial P with integer coefficients such that n divides $P(2^n)$ for every positive integer n . Prove that Big Bird's polynomial must be the zero polynomial.
5. Find all positive integers d with the following property: there exists a polynomial P of degree d with integer coefficients such that $|P(m)| = 1$ for at least $d + 1$ different integers m .
6. Let $\mathbb{Z}[x]$ denote the set of single-variable polynomials in x with integer coefficients. Find all functions $\theta : \mathbb{Z}[x] \rightarrow \mathbb{Z}$ such that for any polynomials $p, q \in \mathbb{Z}[x]$, $\theta(p + 1) = \theta(p) + 1$, and if $\theta(p) \neq 0$ then $\theta(p)$ divides $\theta(p \cdot q)$.
7. Let $P(x)$ and $Q(x)$ be polynomials with integer coefficients. Let $a_n = n! + n$. Show that if $\frac{P(a_n)}{Q(a_n)}$ is an integer for every n , then $\frac{P(n)}{Q(n)}$ is an integer for every integer n such that $Q(n) \neq 0$.
8. Let P be a polynomial with integer coefficients such that $P(0) = 0$ and

$$\gcd(P(0), P(1), P(2), \dots) = 1.$$

Show there are infinitely many n such that

$$\gcd(P(n) - P(0), P(n + 1) - P(1), P(n + 2) - P(2), \dots) = n.$$

9. Find all sequences $a_1, a_2, \dots$ of nonnegative integers such that for all positive integers n , the polynomial

$$1 + x^{a_1} + x^{a_2} + \dots + x^{a_n}$$

has at least one integer root. (Here $x^0 = 1$.)

10. Let $P(x)$ be a polynomial with integer coefficients such that $P(0) = 1$, and let $c > 1$ be an integer. Define $x_0 = 0$ and $x_{i+1} = P(x_i)$ for all integers $i \geq 0$. Show that there are infinitely many positive integers n such that $\gcd(x_n, n + c) = 1$.

5.1.2 Harder

Here are some IMO 2+ problems

1. Let $P(x)$ be a polynomial of degree $n > 1$ with integer coefficients and let k be a positive integer. Consider the polynomial $Q(x) = P(P(\dots P(P(x)) \dots))$, where P occurs k times. Prove that there are at most n integers t such that $Q(t) = t$.

- Let f and g be two nonzero polynomials with integer coefficients and $\deg f > \deg g$. Suppose that for infinitely many primes p the polynomial $pf + g$ has a rational root. Prove that f has a rational root.
- For each positive integer n , let $c(n)$ be the largest real number such that

$$c(n) \leq \left| \frac{f(a) - f(b)}{a - b} \right|$$

for all triples (f, a, b) such that

- f is a polynomial of degree n taking integers to integers, and
- a, b are integers with $f(a) \neq f(b)$.

Find $c(n)$.

- Let $\mathbb{Z}[x]$ denote the set of single-variable polynomials in x with integer coefficients. Find all functions $\theta : \mathbb{Z}[x] \rightarrow \mathbb{Z}[x]$ (i.e. functions taking polynomials to polynomials) such that for any polynomials $p, q \in \mathbb{Z}[x]$, $\theta(p+q) = \theta(p) + \theta(q)$; for any polynomial $p \in \mathbb{Z}[x]$, p has an integer root if and only if $\theta(p)$ does.
- Let $c \geq 1$ be an integer. Define a sequence of positive integers by $a_1 = c$ and

$$a_{n+1} = a_n^3 - 4c \cdot a_n^2 + 5c^2 \cdot a_n + c$$

for all $n \geq 1$. Prove that for each integer $n \geq 2$ there exists a prime number p dividing a_n but none of the numbers $a_1, \dots, a_{n-1}$.

5.2 And Beyond

From here on out you may need to use ideas not covered today. Real polynomial techniques, size comparisons, and anything your creative mind can think of are all on the table!

5.2.1 Part A

Here are some IMO 1- problems to start.

- A repunit is a positive integer whose digits in base 10 are all ones. Find all polynomials f with real coefficients such that if n is a repunit, then so is $f(n)$.
- Let N be the number of positive integers with 10 digits $\overline{d_9 d_8 \cdots d_0}$ in base 10 (where $0 \leq d_i \leq 9$ for all i and $d_9 > 0$) such that the polynomial

$$d_9 x^9 + d_8 x^8 + \cdots + d_1 x + d_0$$

is irreducible in $\mathbb{Q}$. Prove that N is even.

- For any positive integer k , denote the sum of digits of k in its decimal representation by $S(k)$. Find all polynomials $P(x)$ with integer coefficients such that for any positive integer $n \geq 2016$, the integer $P(n)$ is positive and

$$S(P(n)) = P(S(n)).$$

- Determine all nonempty finite sets of positive integers $\{a_1, \dots, a_n\}$ such that $a_1 \cdots a_n$ divides $(x + a_1) \cdots (x + a_n)$ for every positive integer x .

5.2.2 Part B

Some IMO 2+ level problems.

1. Let $P(x)$ be a non-constant polynomial with integer coefficients. Prove that there is no function T from the set of integers into the set of integers such that the number of integers x with $T^n(x) = x$ is equal to $P(n)$ for every $n \geq 1$, where T^n denotes the n -fold application of T .
2. Determine all polynomials f with integer coefficients such that $f(p)$ is a divisor of $2^p - 2$ for every odd prime p .
3. For a polynomial P and a positive integer n , define P_n as the number of positive integer pairs (a, b) such that $a < b \leq n$ and $|P(a)| - |P(b)|$ is divisible by n . Determine all polynomial P with integer coefficients such that $P_n \leq 2021$ for all positive integers n .
4. Find all polynomials f with integer coefficients such that for all positive integers n ,

$$n \text{ divides } \underbrace{f(f(\dots(f(0))\dots))}_{n+1 \text{ } f\text{'s}} - 1.$$

5. Find all integers $n \geq 2$ for which there exists an integer m and a polynomial $P(x)$ with integer coefficients satisfying the following three conditions: $m > 1$ and $\gcd(m, n) = 1$; the numbers $P(0), P^2(0), \dots, P^{m-1}(0)$ are not divisible by n ; and $P^m(0)$ is divisible by n . Here P^k means P applied k times, so $P^1(0) = P(0)$, $P^2(0) = P(P(0))$, etc.
6. Prove there does not exist a monic degree 10 polynomial that is injective mod 101.
7. Let $P(x)$ be a non-constant polynomial with integer coefficients and let n be a positive integer. The sequence $a_0, a_1, \dots$ is defined as follows: $a_0 = n$ and $a_k = P(a_{k-1})$ for all positive integers k . Assume that for every positive integer b the sequence contains a b th power of an integer greater than 1. Show that $P(x)$ is linear.
8. Let $\mathbb{Z}$ denote the set of all integers. Find all polynomials $P(x)$ with integer coefficients that satisfy the following property:
For any infinite sequence $a_1, a_2, \dots$ of integers in which each integer in $\mathbb{Z}$ appears exactly once, there exist indices $i < j$ and an integer k such that $a_i + a_{i+1} + \dots + a_j = P(k)$.
9. Let P be a non-constant polynomial with integer coefficients such that if n is a perfect power, so is $P(n)$. Prove that $P(x) = x$ or P is a perfect power of a polynomial with integer coefficients.
A perfect power is an integer n^k , where $n \in \mathbb{Z}$ and $k \geq 2$. A perfect power of a polynomial is a polynomial $P(x)^k$, where P has integer coefficients and $k \geq 2$.
10. For each integer $k \geq 2$, determine all infinite sequences of positive integers $a_1, a_2, \dots$ for which there exists a polynomial P of the form

$$P(x) = x^k + c_{k-1}x^{k-1} + \dots + c_1x + c_0,$$

where $c_0, c_1, \dots, c_{k-1}$ are non-negative integers, such that

$$P(a_n) = a_{n+1}a_{n+2} \dots a_{n+k}$$

for every integer $n \geq 1$.

11. Find all polynomials $P(x)$ with integer coefficients such that for all real numbers s and t , if $P(s)$ and $P(t)$ are both integers, then $P(st)$ is also an integer.
12. Suppose all the roots of a monic integer polynomial are real and in the interval $(0, 3)$. Prove that these roots must lie in the set $\left\{1, 2, \frac{3 \pm \sqrt{5}}{2}\right\}$.

5.2.3 Part C

Some IMO 3+ level problems. Note that I haven't done a bunch of these.

- Let n be a positive integer and p be a prime. In terms of n and p , find the largest nonnegative integer k for which there exists a polynomial $P(x)$ with integer coefficients satisfying the following conditions:
 - The x^n coefficient of $P(x)$ is 1.
 - p^k divides $P(x)$ for all integers x .
- Let K be the set of all positive integers that do not contain the digit 7 in their base-10 representation. Find all polynomials f with nonnegative integer coefficients such that $f(n) \in K$ whenever $n \in K$.
- An ordered pair (x, y) of integers is a primitive point if the greatest common divisor of x and y is 1. Given a finite set S of primitive points, prove that there exist a positive integer n and integers $a_0, a_1, \dots, a_n$ such that, for each (x, y) in S , we have:

$$a_0x^n + a_1x^{n-1}y + a_2x^{n-2}y^2 + \dots + a_{n-1}xy^{n-1} + a_ny^n = 1.$$

- For a prime p , let $\mathbb{F}_p$ denote the integers modulo p , and let $\mathbb{F}_p[x]$ be the set of polynomials with coefficients in $\mathbb{F}_p$. Find all p for which there exists a quartic polynomial $P(x) \in \mathbb{F}_p[x]$ such that for all integers k , there exists some integer ℓ such that $P(\ell) \equiv k \pmod{p}$. (Note that there are $p^4(p-1)$ quartic polynomials in $\mathbb{F}_p[x]$ in total.)
- We say a nondegenerate triangle whose angles have measures $\theta_1, \theta_2, \theta_3$ is quirky if there exists integers r_1, r_2, r_3 , not all zero, such that

$$r_1\theta_1 + r_2\theta_2 + r_3\theta_3 = 0.$$

Find all integers $n \geq 3$ for which a triangle with side lengths $n-1, n, n+1$ is quirky.

- Let $P(x)$ be a polynomial whose coefficients are all either 0 or 1. Suppose that $P(x)$ can be written as the product of two nonconstant polynomials with integer coefficients. Does it follow that $P(2)$ is a composite integer?
- Find all positive integers n for which there exists a polynomial $P(x) \in \mathbb{Z}[x]$ such that for every positive integer $m \geq 1$, the numbers $P^m(1), \dots, P^m(n)$ leave exactly $\lceil n/2^m \rceil$ distinct remainders when divided by n . (Here, P^m means P applied m times.)
- Find all polynomials $P(x)$ of odd degree d and with integer coefficients satisfying the following property: for each positive integer n , there exists n positive integers $x_1, x_2, \dots, x_n$ such that $\frac{1}{2} < \frac{P(x_i)}{P(x_j)} < 2$ and $\frac{P(x_i)}{P(x_j)}$ is the d -th power of a rational number for every pair of indices i and j with $1 \leq i, j \leq n$.
- A polynomial P with integer coefficients is square-free if it is not expressible in the form $P = Q^2R$, where Q and R are polynomials with integer coefficients and Q is not constant. For a positive integer n , let P_n be the set of polynomials of the form

$$1 + a_1x + a_2x^2 + \dots + a_nx^n$$

with $a_1, a_2, \dots, a_n \in \{0, 1\}$. Prove that there exists an integer N such that for all integers $n \geq N$, more than 99% of the polynomials in P_n are square-free.

6 Solutions

Example 6.1. (Putnam 2006 B1) Show that the curve $x^3 + 3xy + y^3 = 1$ contains only one set of three distinct points, A, B , and C , which are the vertices of an equilateral triangle.

Proof. We have that

$$x^3 + 3xy + y^3 - 1 = \frac{1}{4}(x + y - 1)((x + y + 2)^2 + 3(x - y)^2).$$

Thus the given curve is the union of $(-1, -1)$ and the line $x + y = 1$. The problem is now clear, as one vertex of the equilateral triangle must lie at $(-1, -1)$, and the other two points are uniquely determined by rotating $x + y = 1$ about $(-1, -1)$ by an angle of $\pm 60^\circ$. $\square$

Example 6.2. (CMO 2013 1) Determine all polynomials $P(x)$ with real coefficients such that

$$(x + 1)P(x - 1) - (x - 1)P(x)$$

is a constant polynomial.

Proof. Let $Q(x) = P(x) - P(0)$; then the problem reduces to $(x + 1)Q(x - 1) = (x - 1)Q(x)$. Plugging in $x = \pm 1$ gives us that $0, -1$ are roots of $Q(x)$, so $Q(x) = x(x + 1)R(x)$. Substituting again reduces the equation to $R(x) = R(x - 1)$; thus $R(x) - R(0)$ has infinitely many roots, hence R is a constant. So the solutions are $P(x) = ax(x + 1) + b$ which all work. $\square$

Example 6.3. (HMMT 2017 A6) A polynomial P of degree 2015 satisfies the equation $P(n) = \frac{1}{n^2}$ for $n = 1, 2, \dots, 2016$. Find $2017P(2017)$.

Proof. Consider $Q(x) = x^2P(x) - 1$. Then $Q(x)$ must factor as $Q(x) = c(x - r)\prod_{i=1}^{2016}(x - i)$ for some c, r . Since $Q(0) = -1$ we find that $cr(2016)! = 1$. Since the x^1 coefficient of $Q(x)$ is zero we find that the reciprocals of the roots of Q sum to zero, hence $\frac{1}{r} = -\sum_{i=1}^{2016} \frac{1}{i}$. Thus

$$\begin{aligned} 2017^2P(2017) &= Q(2017) + 1 \\ &= c(2017 - r)(2016)! + 1 \\ &= \frac{2017 - r}{r} + 1 \\ &= \frac{2017}{r}. \\ 2017P(2017) &= \frac{1}{r} \\ &= -\sum_{i=1}^{2016} \frac{1}{i}. \end{aligned}$$

$\square$

Example 6.4. (Putnam 2007 B4) Let n be a positive integer. Find the number of pairs P, Q of polynomials with real coefficients such that

$$(P(X))^2 + (Q(X))^2 = X^{2n} + 1$$

and $\deg P < \deg Q$.

Proof. Let $R(x) = Q(x) + iP(x)$. Then $x^{2n} + 1 = R(x)\overline{R(x)}$. Note that by the degree condition, Q (hence R) both must have degree n and have leading coefficient ± 1 . So it remains to distribute the roots of $x^{2n} + 1$ to $R(x)$ and $\overline{R(x)}$. But if R has a root z then $\overline{R}$ must have the root $\bar{z}$; the roots of $x^{2n} + 1$ come in n complex conjugate pairs, so there are 2^n ways to distribute the roots to R and $\overline{R}$. Choosing the sign of the leading coefficient doubles the possibilities for R (and hence P, Q), so there are 2^{n+1} pairs (P, Q) . $\square$

Example 6.5. (CMO 2016 3) Find all polynomials $P(x)$ with integer coefficients such that $P(P(n) + n)$ is a prime number for infinitely many integers n .

Proof. Any constant $P(x)$ always equal to a prime number works, so assume P is nonconstant. Observe $P(n)|P(P(n) + n) - P(n)$ so $P(n)|P(P(n) + n)$. Thus $P(n) \in \{\pm 1\}$ for infinitely many n , or $|P(n)| = |P(P(n) + n)|$ for infinitely many n . In the former case $P(n)$ must be one of 1 or -1 infinitely often, so P is constant. In the latter case, by comparing degrees we must have $\deg P = 1$; by comparing leading coefficients we find that $P(x) = c - 2x$ for some integer c . Then $P(P(n) + n) = 2n - c$, so $P(x) = c - 2x$ works iff c is odd. Hence the solutions are P constant to a prime or $P(x) = (2n + 1) - 2x$. $\square$

Example 6.6. (ELMO 2023 1) Let m be a positive integer. Find, in terms of m , all polynomials $P(x)$ with integer coefficients such that for every integer n , there exists an integer k such that $P(k) = n^m$.

Proof. By setting $n = 0$ we find that P has an integer root at some c . By replacing $P(x)$ with $P(x - c)$ we may assume $P(0) = 0$, so $P(x) = xQ(x)$ for some integer polynomial Q . Now for any prime p , there exists an integer k such that $kQ(k) = p^m$; hence $k = \pm p^a$ and $Q(k) = \pm p^{m-a}$ for some integer $1 \leq a \leq m$. By infinite pigeonhole some choice of a (and signs) occurs infinitely many often. In that case, since $Q(x) = \Theta(x^{\deg Q})$, we must have that $a|m$. Moreover, depending on the choice of signs either $Q(x) - x^{m/a-1}$ or $Q(x) + x^{m/a-1}$ must have infinitely many roots. Hence $Q(x) = \pm x^i$ for some $i+1|m$, and (after taking out the rescaling) $P(x) = \pm(x - c)^d$ for some integer c and some $d|m$. All such P work so we are done. $\square$

Example 6.7. (Taiwan TST 2014 Round 1) For a fixed integer k , determine all polynomials $f(x)$ with integer coefficients such that $f(n)$ divides $(n!)^k$ for every positive integer n .

Proof. Let $f(x) = x^l g(x)$ for some $g(x)$ not divisible by x ; by investigating the case $x = p$ we find that $p^l|(p!)^k$ and so $l \leq k$. So the given condition implies that $g(n)|n^{k-l}((n-1)!)^k$ for all positive n . Now, suppose a prime p divides some $g(n)$. By taking remainders, p also divides $g(m)$, where m is the unique integer satisfying $1 \leq m \leq p$ and $m \equiv n \pmod{p}$. But then $p|m^{k-l}((m-1)!)^k$, so $p \leq m$. This is only possible if $p = m$, in which case $p|g(p)$ and so $p|g(0)$. If g is nonconstant, then infinitely many such P exist, implying that $g(0) = 0$, a contradiction. So g is a constant, and so $f(n) = cn^l$ for some integer c and $l \leq k$. Plugging in $n = 1$ now gives us that $c \in \{\pm 1\}$. $\square$

Example 6.8. (2011 N6) Let $P(x)$ and $Q(x)$ be two polynomials with integer coefficients, such that no nonconstant polynomial with rational coefficients divides both $P(x)$ and $Q(x)$. Suppose that for every positive integer n the integers $P(n)$ and $Q(n)$ are positive, and $2^{Q(n)} - 1$ divides $3^{P(n)} - 1$. Prove that $Q(x)$ is a constant polynomial.

Proof. First by Bézout there exists integer polynomials $A(x), B(x)$, and a constant integer c such that $A(x)P(x) + B(x)Q(x) = c$. Now, the given condition implies that for any prime p ,

$$\text{ord}_p(2)|Q(n) \implies \text{ord}_p(3)|P(n)$$

where ord denotes order modulo p . Now, suppose FTSOC that $Q(x)$ is nonconstant. Consider some prime q which divides $Q(n)$ for some n ; infinitely many such q exist by Schur, and for all sufficiently large choices of q , there exists a prime p with $\text{ord}_p(2) = q$ by Zsigmondy. The above implies that $q|Q(m)$ implies that $x_q = \text{ord}_p(3)|P(m)$.

Now, suppose for infinitely many of our q , $q|x_q$ is true. Then $q|P(m), Q(m)$ in this case, and so $q|c$, which is a contradiction. Thus infinitely many of our q must satisfy that $q \nmid x_q$. In this case we have that $q|Q(n + kq)$ for all k , hence x_q divides $P(n + kq)$ for all k as well. But x_q also divides $P(n + kq + lx_q)$ for all l ; since $\gcd(x_q, q) = 1$, we have that x_q divides $P(0)$. Now, as we increase q , the sequences of p s and $\text{ord}_p(3) = x_q$ s we get as a result must both be unbounded (this is left as an exercise to the reader). Thus it is also impossible for infinitely many x_q to divide $P(0)$, a contradiction, as desired. $\square$

Example 6.9. Given any two distinct positive integers a, b , prove that $\sqrt{a} + \sqrt{b}$ is an algebraic integer.

Proof. $p(x) = (x - \sqrt{a} - \sqrt{b})(x - \sqrt{a} + \sqrt{b})(x + \sqrt{a} - \sqrt{b})(x + \sqrt{a} + \sqrt{b})$ works. $\square$

Example 6.10. (Niven) Consider a rational number q . Suppose $\cos(q\pi)$ is rational; then $\cos(q\pi) \in \{\pm 1, \pm 1/2, 0\}$.

Proof. Let $z = e^{\pi i q}$; then z and $\bar{z} = z^{-1}$ are both roots of unity and hence algebraic integers. Thus $2\cos(q\pi) = z + \bar{z}$ is also an algebraic integer, but it's also rational. Hence it must be an integer; it's at most 2 in absolute value, hence the conclusion. $\square$

Example 6.11. (Black Mop 2010) Consider $n > 2$ lamps placed at the vertices of a regular n -gon. Each move we may choose, for some divisor $d > 1$ of n , d lamps forming a regular d -gon, and toggle all of these lamps once. Suppose the lamps begin with one on and all others off. Is it possible to turn all lamps off after a finite number of moves?

Proof. The answer is no; suppose such a sequence of moves exists FTSOC. Let $\omega = e^{2\pi i/n}$, and label the lamps clockwise from 0 to $n - 1$, where the initially on lamp is labelled 0. Let c_i be the number of toggles lamp i has received. Then

$$\sum_i c_i \omega^i$$

is invariant after each move. Thus at the end we have that

$$\sum_i c_i \omega^i = 0$$

and all c_i except c_0 are even. Letting $d_i = c_i/2$ for $i \neq 0$ and $d_0 = (c_0 + 1)/2$, we find that after our sequence of moves,

$$\begin{aligned} 2 \sum_i d_i \omega^i &= 1, \\ \sum_i d_i \omega^i &= \frac{1}{2}. \end{aligned}$$

But the LHS is an algebraic integer while the RHS is not, a contradiction. $\square$